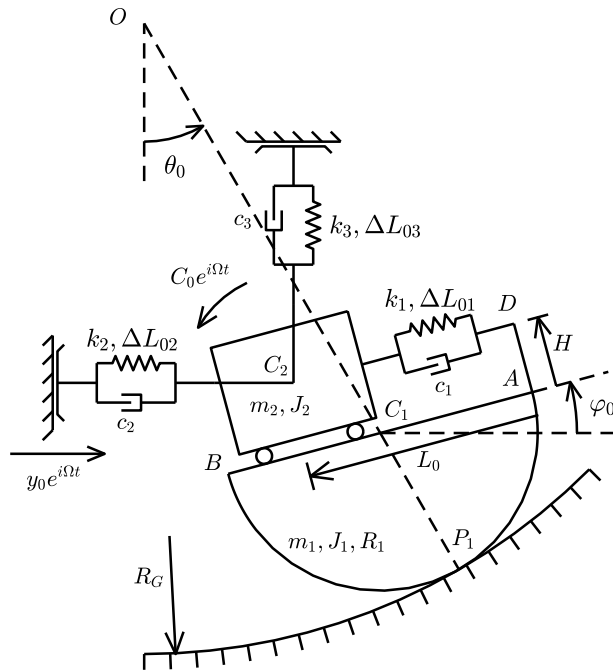


Dynamics of Mechanical Systems

June 10th, 2021



$m_1 = 10 \text{ kg}$	$c_1 = 3 \text{ Ns/m}$
$m_2 = 10 \text{ kg}$	$c_2 = 3 \text{ Ns/m}$
$J_1 = 0.15 \text{ kgm}^2$	$c_3 = 3 \text{ Ns/m}$
$J_2 = 0.08 \text{ kgm}^2$	$\theta_0 = \pi/6$
$R_1 = 0.25 \text{ m}$	$\varphi_0 = \pi/12$
$R_G = 1.0 \text{ m}$	
$L_0 = 0.35 \text{ m}$	
$H = 0.1 \text{ m}$	
$k_1 = 3000 \text{ N/m}$	
$k_2 = 3000 \text{ N/m}$	
$k_3 = 3000 \text{ N/m}$	
$\Delta L_{01} = -0.0041 \text{ m}$	
$\Delta L_{02} = -0.0076 \text{ m}$	
$\Delta L_{03} = 0.02 \text{ m}$	

The mechanical system above lies in the vertical plane and is shown in its equilibrium position. The half disk rolls without sliding and its center of mass can be considered coincident with the geometrical center C_1 .

The students are requested to:

0. Write on top of each page their surname, name, ID number and SNxxx code (see below)
1. Write the linearized equations of motion around the given equilibrium configuration
2. Compute the natural frequencies and modal shapes;
3. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic torque $C_0 e^{i\Omega t}$ to the mass m_2 . Output: the rotation of the mass m_2 and the horizontal component of the displacement of point C_2 . Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your person ID number (e.g. for Bruni Stefano 12345678: BS6781.fig and BS6782.fig)
4. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic constraint displacement $y_0 e^{i\Omega t}$. Output: the horizontal component of the displacement of the point C_1 and the elastic and damping force of the spring and damper system (k_3, c_3). Save the graphical results as SNxxx3.fig and SNxxx4.fig
5. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic constraint displacement $y_0 e^{i\Omega t}$. Output: the normal and the tangential dynamic components of the contact force between the half disk and the circular guide. Save the graphical results as SNxxx3.fig and SNxxx4.fig

The answer to points 2 – 5 shall be provided by delivering the MATLAB script and .fig files as a compressed folder uploaded on BeeP, see instructions below.

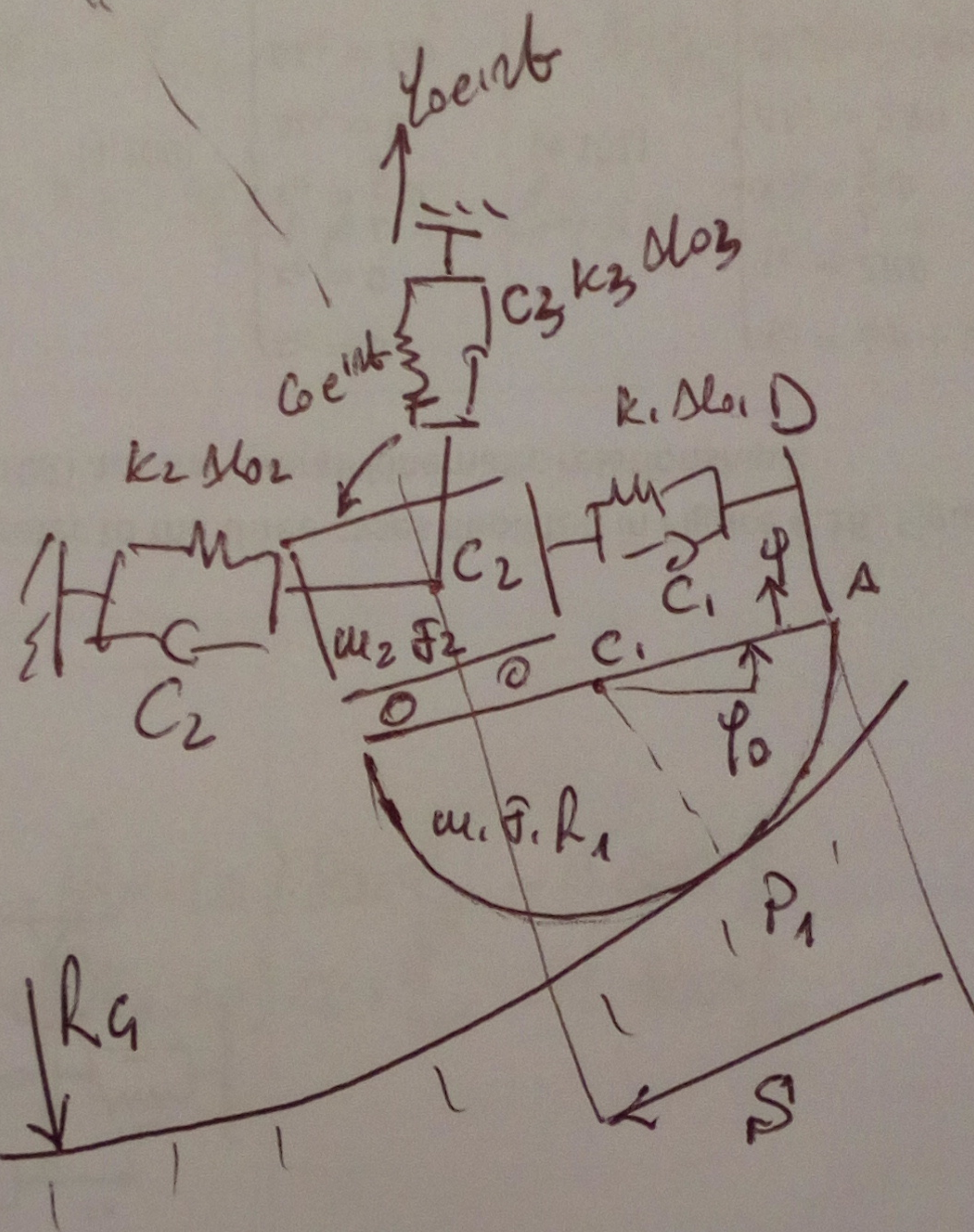
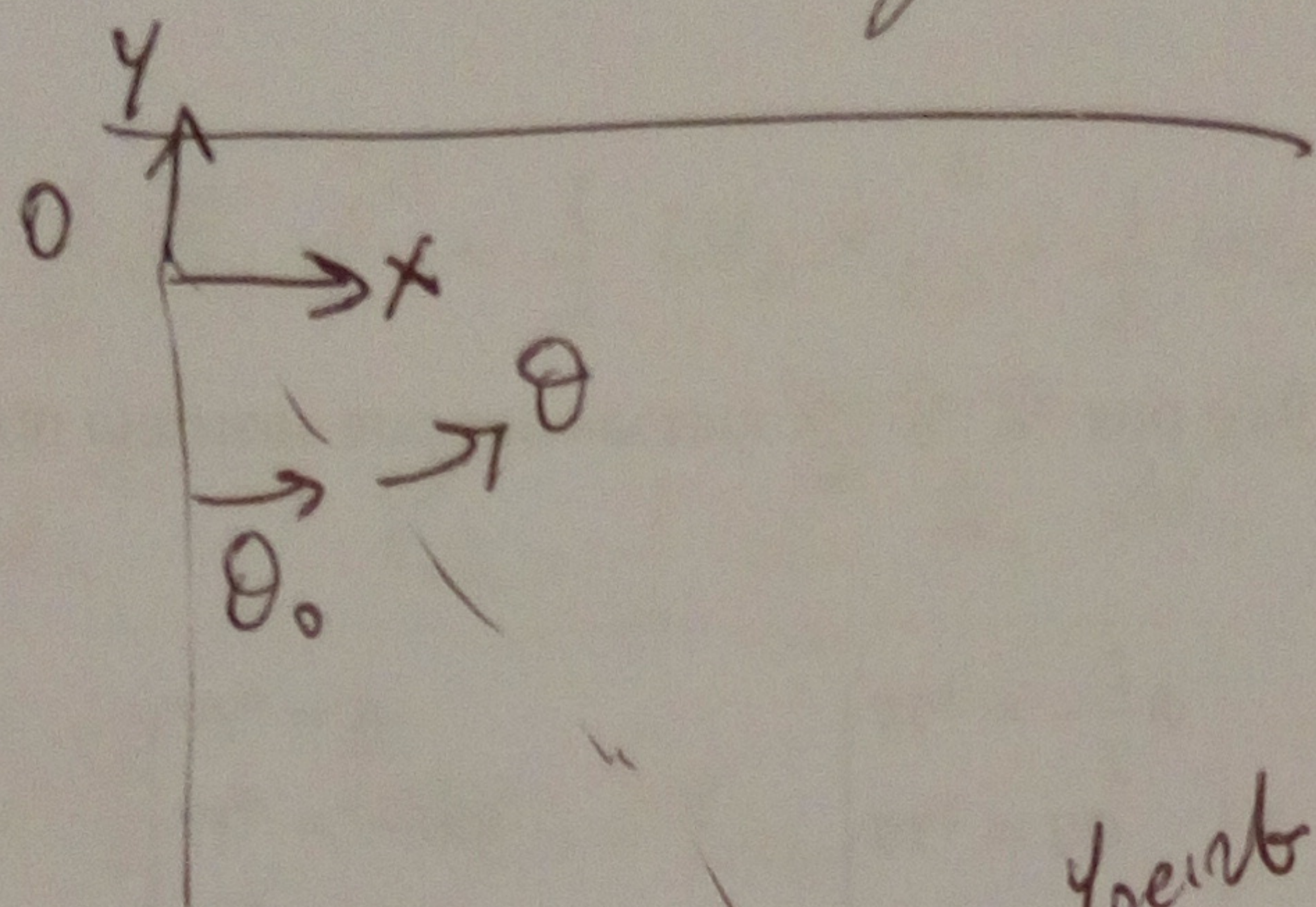
Please be aware that the mark will be lowered by two points if:

- pdf files without the SNxxx code on top of the first page;
- the name of the zip file different from SNxxx.zip;
- the Beep field *Title* filled not accordingly to the indications given below at point 7.

Note: Please collect the Matlab's script file relative to the numerical solution of the exercise and save it with name SNxxx.m, with S the first letter of the surname, N the first letter of the name and xxx the last three digits of person ID number / codice persona (e.g. for Bruni Stefano 12345678: BS678.m). It is mandatory that all of the files (fig e .m) are collected in one single compressed folder (in .zip or.rar format) named SNxxx.zip (or SNxxx.rar), which has to be uploaded on Beep. To upload the sipped folder:

1. Connect to the site <https://beep.metid.polimi.it> and enter the portal using your credentials;
3. Select section **Homework** of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select the folder **Test June 10th 2021 - standard**
5. Click on button Add and select Single document;
6. Browse the compressed file SNxxx.zip using the button Browse... (or Sfogliala...)
7. Fill the "Title" field with your code SNxxx (e.g. for Bruni Stefano 123456: BS456);
8. Confirm the procedure clicking the button Publish.

Written Test June 10th 2021
- Standard problem -



2 dof.

$$\underline{x} = \begin{Bmatrix} \phi \\ s \end{Bmatrix} \quad \underline{x}_0 = \begin{Bmatrix} \phi_0 \\ L_0 \end{Bmatrix}$$

component of the

ϕ : dynamic rotation of the half disk

s : ~~posit~~ relative position of the mass u_2 with respect D.

$$R = R_g - r_1$$

$$R\dot{\theta} = -r_1\dot{\phi} \rightarrow \dot{\theta} = -\frac{r_1}{R}\dot{\phi} \rightarrow \theta = -\frac{r_1}{R}\phi$$

$$x_{c1} = R \sin(\theta_0 + \theta)$$

$$y_{c1} = -R \cos(\theta_0 + \theta)$$

$$x_A = R \sin(\theta_0 + \theta) + r_1 \cos(\phi_0 + \phi)$$

$$y_A = -R \cos(\theta_0 + \theta) + r_1 \sin(\phi_0 + \phi)$$

$$x_{c2} = R \sin(\theta_0 + \theta) + (R_1 - S) \cos(\phi_0 + \phi) - H \sin(\phi_0 + \phi)$$

$$y_{c2} = -R \cos(\theta_0 + \theta) + (R_1 - S) \sin(\phi_0 + \phi) + H \cos(\phi_0 + \phi)$$

$$T = \frac{1}{2} m_1 \dot{x}_{c1}^2 + \frac{1}{2} J_1 \dot{\phi}^2 + \frac{1}{2} m_2 \dot{x}_{c2}^2 + \frac{1}{2} m_2 \dot{y}_{c2}^2 + \frac{1}{2} J_2 \dot{\phi}^2$$

$$D = \frac{1}{2} \sum_{i=1}^3 c_i \Delta l_i^2 ; V_E = \frac{1}{2} \sum_{i=1}^3 k_i \Delta l_i^2$$

$$\delta W = C_0 \sin \theta \delta \phi + F_{\phi y3} \delta y_{c2} ; F_{\phi y3} = k_3 y + C_3 \dot{y}$$

$$\Delta l_1 = S - l_0 \quad \Delta l_2 = x_c - x_{c2p} \quad \Delta l_3 = -y_{c2} + y_{c2p}$$

first derivatives $\rightarrow$

$$[\Delta u] = \begin{bmatrix} R_1 & & & & \\ 1 & & & & \\ -R_1 \cos \theta_0 - (R_1 - l_0) \sin \phi_0 - H \cos \phi_0 & \phi & & & \\ -R_1 \sin \theta_0 + (R_1 - l_0) \cos \phi_0 - H \sin \phi_0 & -\cos \phi_0 & & & \\ & -\sin \phi_0 & & & \\ & & 1 & & \\ & & & \phi & \end{bmatrix} \begin{matrix} \delta \phi \\ \theta \\ x_{c2} \\ y_{c2} \\ \phi \end{matrix}$$

$$[\Delta u] = [\Delta c] = \begin{bmatrix} 0 & & & & \\ -R_1 \cos \theta_0 - (R_1 - l_0) \sin \phi_0 - H \cos \phi_0 & 1 & & & \\ R_1 \sin \theta_0 - (R_1 - l_0) \cos \phi_0 + H \sin \phi_0 & -\cos \phi_0 & & & \\ & \sin \phi_0 & & & \end{bmatrix} \begin{matrix} \Delta l_1 \\ \Delta l_2 \\ \Delta l_3 \end{matrix}$$

$$\Delta q = \begin{bmatrix} 1 & & & & \\ -R_1 \cos \theta_0 - (R_1 - l_0) \sin \phi_0 - H \cos \phi_0 & \phi & & & \\ & -\cos \phi_0 & & & \end{bmatrix} \begin{matrix} C_0 \\ F_{\phi y3} \end{matrix}$$

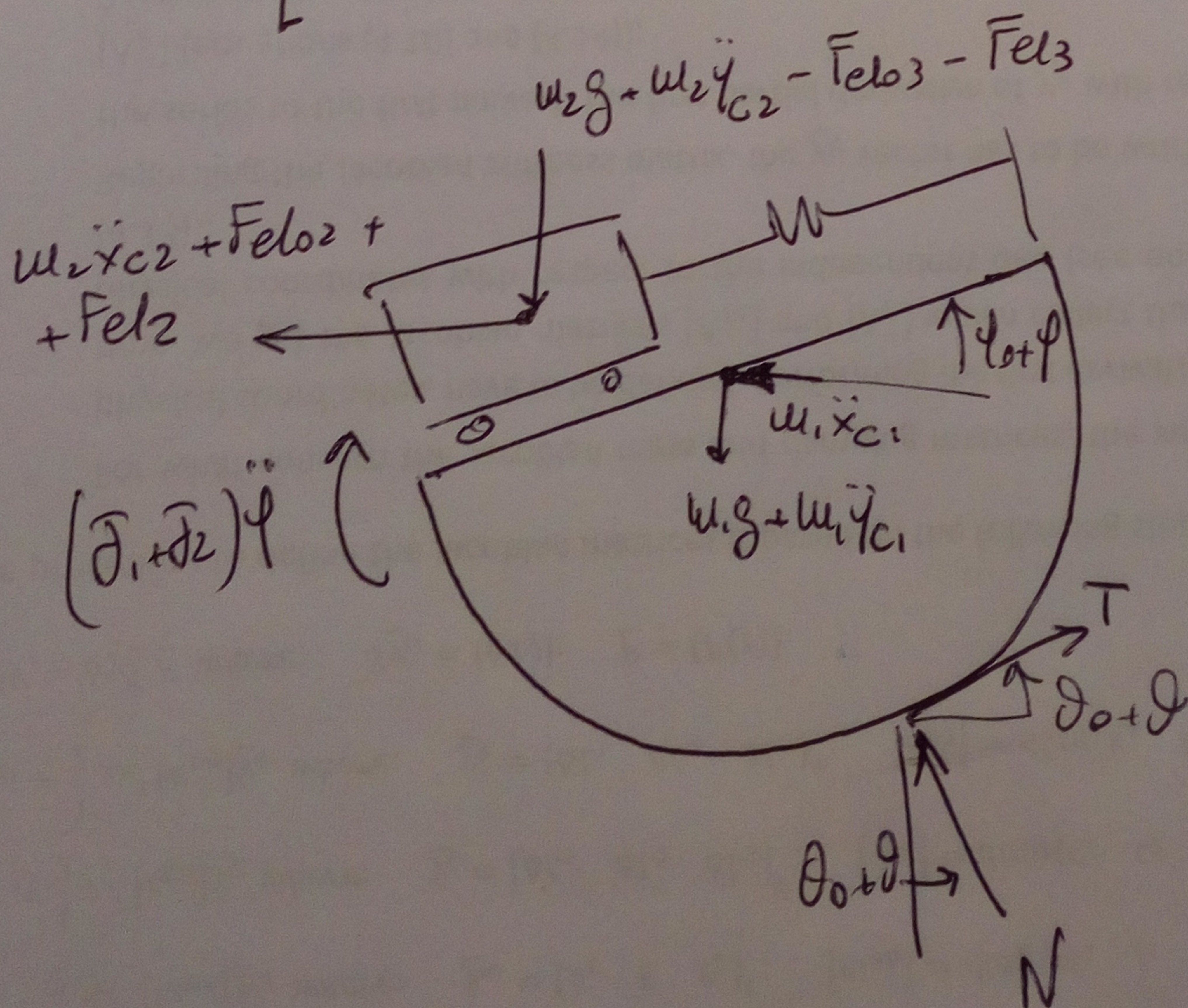
Second derivatives $\rightarrow$

$$[K_{EL, \pi_2}] = k_2 \Delta l_2 \begin{bmatrix} -\frac{R_1^2}{R} \sin \theta_0 - (R_1 - l_0) \cos \phi_0 + H \sin \phi_0 & \sin \phi_0 \\ \sin \phi_0 & \rho \end{bmatrix}$$

$$[K_{EL, \pi_3}] = k_3 \Delta l_3 \begin{bmatrix} -\frac{R_1^2}{R} \cos \theta_0 + (R_1 - l_0) \sin \phi_0 + H \cos \phi_0 & \cos \phi_0 \\ \cos \phi_0 & \phi \end{bmatrix}$$

$$[K_{s1}] = m_1 g \begin{bmatrix} \frac{R_1^2}{R} \cos \theta_0 & \phi \\ \phi & \phi \end{bmatrix}$$

$$[K_{s2}] = m_2 g \begin{bmatrix} \frac{R_1^2}{R} \cos \theta_0 - (R_1 - l_0) \sin \phi_0 - H \cos \phi_0 & -\cos \phi_0 \\ -\cos \phi_0 & \phi \end{bmatrix}$$



$$\begin{cases} T \cos(\theta_0 + \theta) - N \sin(\theta_0 + \theta) - m_1 \ddot{x}_{c1} - m_2 \ddot{x}_{c2} - F_{el2} - F_{el2} = 0 \\ T \sin(\theta_0 + \theta) + N \cos(\theta_0 + \theta) - m_1 g - m_1 \ddot{y}_{c1} - m_2 g - m_2 \ddot{y}_{c2} + F_{el3} + F_{el3} = 0 \end{cases}$$

$$\begin{cases} T = T_{ST} + T_d \\ N = N_{ST} + N_d \end{cases}$$

considering only the dynamic components:

$$\begin{cases} T_d \cos \theta_0 - N_d \sin \theta_0 = m_1 \ddot{x}_{c1} + m_2 \ddot{x}_{c2} + F_{el2} \\ T_d \sin \theta_0 + N_d \cos \theta_0 = m_1 \ddot{y}_{c1} + m_2 \ddot{y}_{c2} - F_{el3} \end{cases}$$

$$\begin{bmatrix} \cos \theta_0 & -\sin \theta_0 \\ \sin \theta_0 & \cos \theta_0 \end{bmatrix} \begin{Bmatrix} T_d \\ N_d \end{Bmatrix} = \begin{Bmatrix} m_1 \ddot{x}_{c1} + m_2 \ddot{x}_{c2} + F_{el2} \\ m_1 \ddot{y}_{c1} + m_2 \ddot{y}_{c2} - F_{el3} \end{Bmatrix}$$